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Solvent-Organization and Rate-Regulation of a Menshutkin Reaction by Oriented-External Electric Fields are Revealed by Combined MD and QM/MM Calculations

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Abstract

When and how do external electric fields (EEF) lead to catalysis in the presence of a (polar or nonpolar) solvent? This is the question that is addressed here using a combination of molecular-dynamics (MD) simulations, quantum-mechanical / molecular-mechanical with EEF, and quantum-mechanical / (local) electric-field calculations. The manuscript focuses on a model-reaction, the Menshutkin reaction between CH_3I and pyridine in three solvents of varying polarity. Using MD simulations, we find that the EEF causes the solvent to undergo organization; *the solvent molecules gradually align with the applied field as the field strength increases*. The collective orientation of the solvent-molecules modifies the electrostatic environment around the Menshutkin species and induces a global electric field pointing in the opposite direction of the applied EEF. The combination of these two entangled effects leads to partial-or-complete screening of the EEF, with the extent of screening being proportional to the polarity/polarizability of the solvent. *Nevertheless, we find that the catalysis of the Menshutkin reaction inevitably emerges once the EEF exceeds the opposing field of the organizing solvent, i.e. once polarization of the Menshutkin complex is observed to set in*. Overall, our analysis

provides a lucid and pictorial interpretation of the behavior of solutions in the presence of EEFs and indicates that EEF-mediated catalysis should in principle be feasible in bulk setups, especially for nonpolar and mildly polar solvents. By application of the charge-transfer paradigm, it is shown that the emergence of OEEF catalysis in solution can be generalized to other reactions as well.

Introduction

It has been well established by now that external-electric fields (EEF) can affect chemical reactivity and structure.^{1,2} By use of a single-molecule scanning-tunneling microscopy (STM) experimental setup, Aragonés *et al.* offered an initial proof-of-principle that a Diels-Alder reactions, which involves concerted two C-C bond making, can be catalyzed by EEFs.³ Since then, multiple groups have used similar setups to probe EEF-mediated reactivity.^{4, 5} Very recently, Venkataraman and co-workers demonstrated that the *cis/trans* isomerization of cumulenes can be catalyzed and directed in an STM experiment.⁶ Other experimental approaches exist which may in principle harness electrostatics in chemical reactivity in bulk, involving surface charging,⁷ or the generation of interfacial electric fields in non-Faradic electrochemical cells.⁸

Alternatively, the use of so-called local electric fields (LEF) have been explored.^{1b,c} LEFs are electric fields generated by charged functional groups or dipoles at distant sites of a reactant or catalyst, which can potentially influence the reactivity exhibited at the active site/reactive center.^{9,10,11,12}

Electric-field mediated catalysis would become practically useful, upon development of scalable and cost-effective methods. Most of the techniques and approaches mentioned above are either not inherently scalable, or they put limitations on the structural diversity of the reactants involved in the chemical system under consideration. Some reports on esterification

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3 reactions in solvents have recently suggested that pulsed electric fields (PEF) could impart
4 electrostatic catalysis in a continuous-flow reactor setup.¹³ While such an approach would in
5 principle be scalable, its broad-range effectiveness has not been demonstrated yet. Importantly,
6 a clear understanding of the effect of the presence of (bulk) solvent on the catalysis effected by
7 EEFs is not yet available, so that the exact mechanism whereby PEFs – or more generally
8 speaking, any EEF in solution – could affect chemical reactivity is currently still clouded.
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11 The common understanding is that solvent molecules collectively screen any exerted
12 EEF, so that in a bulk situation, the global field effect should become – either partially or
13 completely – canceled out. The ability for a solvent to screen an external field is understood to
14 depend on the dielectric constant and polarizability of the solvent, i.e. on its polarity.¹⁴
15 Additionally, it is known that intense EEFs may cause solvents to evaporate, and bring about
16 the breakdown of the solvent molecules or reactants, thus generating currents and many
17 byproducts.⁸ These phenomena are often lumped together and are collectively described as
18 “dielectric breakdown”.¹⁵ Despite these apparent limitations, there have been several
19 experimental treatments of reactions in solvents under the influence of an EEF by Kanan,⁸
20 Matile,⁷ Ciampi and Coote,¹⁶ and even the STM experiments we have referred to above are
21 carried out in (nonpolar) solvents.^{3,4,5} In the STM study by Venkataraman *et al.*, the influence
22 of different solvents on the catalytic activity exerted by an applied EEF was compared directly
23 and a significant decrease of catalysis for polar solvents compared to nonpolar ones was found.⁶
24 Furthermore, Cassone *et al.* used state-of-the-art Car-Parrinello MD/DFT (CPMD/DFT)¹⁷
25 calculations to simulate various reactions in water and in alcohol under the influence of an
26 external field and reported a variety of interesting results.¹⁸
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29 The present manuscript is focused on shedding some more light on the effect of the
30 presence of solvent on EEF-mediated catalysis. To this end, we have decided to investigate a
31 synthetically useful reaction, the Menshutkin reaction.¹⁹ The reaction system, pyridine and
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methyl iodide, as well as the convention used for the electric field direction, are shown in Fig.

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a)



b)

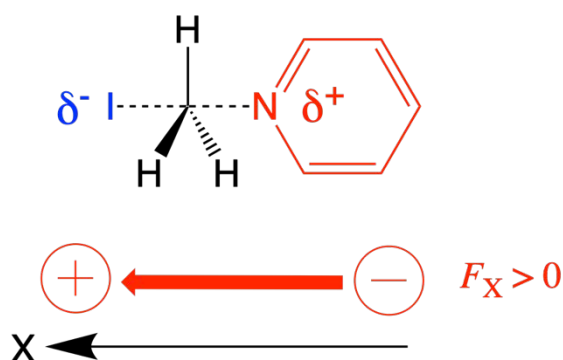


Figure 1: a) A Schematic representation of the Menshutkin reaction of Pyridine with Methyl Iodide. b) The associated transition state (TS) geometry, wherein the reaction axis is indicated as a dashed line, and the convention for a *positively oriented* EEF vector along the reaction axis (F_X), which induces catalysis in the gas-phase. Note that throughout this work, we employ the Gaussian convention for the electric field direction, i.e. directed from negative to positive.

It has been understood for quite some time that exposing the reaction complex associated to the Menshutkin reaction in the gas-phase to an oriented external electric field with $F_X > 0$ along the reaction axis (cf. Fig. 1b), induces catalysis, whereas flipping the direction of the field induces inhibition.^{1,20} The detailed mechanism behind the catalytic activity of EEFs for this type of gas-phase reactivity has been discussed at length on many different occasions, cf. Ref. 1a and 1c for a detailed valence bond (VB) treatment for example. In short, one could say that the EEF induces polarization of the reacting species, in the direction which facilitates

the “flow” of the electrons throughout the transformation from reactant to the two ionic products, and thereby lowers the reaction barrier and increases the thermodynamic driving force by stabilizing the transition state (TS) and the products. How will the presence of solvent molecules affect this catalysis? That is the question we try to address below.

Throughout our discussions, we will mainly focus on our findings for the prototypical polar solvent CH₃CN ($\epsilon = 36.6$), but similar trends are observed for the other solvents considered, i.e. chloroform ($\epsilon = 4.8$) and acetone ($\epsilon = 21.0$), *vide infra*.²¹

The reaction depicted in Fig. 1a will be studied through a combination of molecular dynamic (MD) simulations under the influence of an EEF (F_X , Fig. 1) of a varying field strength, followed by quantum mechanical (QM) /Molecular mechanical (MM)²² and quantum mechanical (QM) /electric field (EF)^{23,24} calculations. The conceptual picture emerging from these calculations provides a lucid interpretation and pictorial understanding of the ubiquitous solvent screening effect and the behavior of polar solutions in the presence of external electric fields.

Methodology

Molecular dynamics of the solvated systems: All MD simulations with and without an EEF were carried out using the AMBER package.²⁵ Detailed information about the construction of the solvent boxes for the different solvents, introduction of the Menshutkin complex and the setup of the MD simulations up to the equilibration step can be found in Section S1 in the Supporting Information (SI). To simulate the effect of electric fields on the full solvated system, an EEF (F_X) was applied in the direction of the reaction coordinate, i.e. in the direction of the N---C---I reaction axis, which is aligned to the +X direction in the modeled system, at different field strengths. We applied F_X values of 0.02, 0.03, 0.035, 0.04, 0.05, 0.06, 0.07, 0.1, 0.15, and 0.2 V/Å using the option “external electric”, implemented in the Amber MD package. At each

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3 F_X value, we reran the MD simulation. All these production dynamics were each carried out
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5 for 10 ns. Since the final few nanoseconds of the simulations revealed no conformational
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7 changes in the solvent, the chosen simulation timeframe was considered adequate/sufficient.
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9 Entropies of all simulated complexes were calculated using the Cpptraj module of Amber MD
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11 package which makes use of the quasi harmonic approximation method. Let us note that while
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13 our simulations do not include solvent polarization, still this polarization will definitely occur
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15 in the solvent molecules, based on the extensive calculations of this effect in molecules
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17 subjected to EEFs.^{1,2}
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21 *Calculation of the gas-phase reaction profile:* The gas-phase PES associated with the
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23 Menshutkin reaction was determined by use of Gaussian09.²⁶ For geometry optimization and
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25 frequency calculations, we used the hybrid B3LYP²⁷ functional and a basis set B1, consisting
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27 of Def2-SVP for all atoms. The energies were further corrected with the large all-electron basis
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29 set Def2-TZVP, labeled as B2. The zero-point energy (ZPE) was calculated for all species, and
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31 all the final energies are reported as UB3LYP/B2+ZPE. Entropic contributions were not taken
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33 into account in the constructed potential energy surface; as discussed in Section S3 in the SI,
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35 the effect of the entropy on the EEF-mediated catalysis (*vide infra*) can be expected to be minor.
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40 *The electrostatic effect of the solvent environment on the reaction:* Two distinct
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42 calculations were performed on a representative snapshot of the MD simulation in the absence
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44 of an electric field, to determine the effect of the solvent environment on the Menshutkin
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46 reaction and assess the extent to which this effect is electrostatic in nature. This representative
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48 snapshot was chosen from the clustering of the MD trajectories, which provides the most
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50 populated trajectories along with the representative snapshot for each cluster. This protocol is
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52 statistically more accurate relative to the stochastically chosen snapshot.²⁸
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56 First, a QM/MM calculation was executed such that the reactants, pyridine and methyl
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58 iodide, were treated at a QM level-of-theory, while all solvent molecules were kept as “active
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MM atoms". This QM/MM calculation was performed using ChemShell,^{29,30} combining Turbomole³¹ for the QM part and DL_POLY³² for the MM part using the Amber force field. The electronic embedding scheme was used to account for the polarizing effect of the solvent environment on the QM region. During the geometry optimizations, the QM region was treated by the hybrid B3LYP functional with two basis sets, in accordance with the gas-phase calculations described above. As such, we used basis set B1 for geometry optimization and frequency calculations and corrected the resulting energies with the B2 basis set. The ZPE was again calculated for all species, and all the final energies are reported as UB3LYP/B2+ZPE.

Subsequently, the point-charge distribution associated to the solvent molecules were extracted and inserted into Gaussian 09,²⁶ to carry out a single-point QM/EF calculation, where EF represents the point charge distribution of the solvent molecules. UB3LYP/B2+ZPE was used here, in order to test the consistency of QM/EF with the QM/MM calculation. Comparison of the reaction barriers emerging from these two calculations then enabled the confirmation of the electrostatic nature of the solvent effects (*vide infra*).

The screening effect of the ordered solvent environment: To assess the magnitude of the induced-electric field exerted by the ordered solvent on the reaction axis of the reaction complex, the induced field was quantified using the in-house developed TITAN code.²⁴ First, the ordered solvent environment, extracted from the MD simulation at 0.2 V/Å, was translated into a charge distribution, in which the individual atoms were assigned by TITAN the same charge parameters as in the MD simulations. Subsequently, the oriented electric field exerted along the N---C----I axis was quantified through a calculation of the individual field vectors associated to each of the point charges in this distribution at different positions along this axis according to Coulomb's Law.²⁴

The effect of an applied EEF on the Menshutkin reaction in solvent environments: We performed QM/MM calculation of the solvated Menshutkin-reaction system under the

influence of an EEF exerted along the X-axis, in order to assess the effect of the oriented EEF on the potential energy surface (PES) associated to the reaction. A representative snapshot obtained from the MD simulation at 0.2 V/Å was selected as the input geometry for the simulation box. Since DL-POLY does not contain a keyword to take into account the presence of an EEF in QM/MM calculations, a pair of charged circular plates³³ was generated to mimic such a field with the help of the TITAN code.²⁴ The generated plates consisted of 33 circular rings of charged dummy atoms, with the spacing between each consecutive ring set to 2.8 Å. Both plates were placed 50 Å away from the C-atom in the middle of the N---C---I axis of the reaction complex on the N- and I-ends. The individual charges were set, so as to generate a uniform field strength of 0.5 V/Å inside the solvated system.

Subsequently, we subdivided the solvent box into two parts: all solvent molecules within 6 Å around the reaction complex were taken as the “active” MM region. The outer layer of ordered solvent was designated as “inactive” during the QM/MM calculation; hence its positions remained frozen (cf. Section S2 in the Supporting Information). The reproducibility of the so calculated energy barriers was tested by rotating the plates by 90° and redoing the QM/MM calculations (see section S10 in the SI).

Note that since all solvent molecules are treated at the MM level-of-theory in this final QM/MM calculation, the charges on the respective atoms and the bond lengths remain at their fixed parametrized values throughout the entire simulation. This in turn, does not account for the gradual polarization of the solvent molecules under the influence of an applied EEF, and the consequent increase of the counteracting, global induced solvent field, which is expected to attenuate the catalytic effect of the applied EEF (*vide infra*). Similarly, the inability to describe the polarization taking place in the solvent also implies we are unable to calculate “dielectric breakdown” in our present simulations as the electric field strength is increased.

Results and discussions

The Menshutkin reaction in solution without EEF

Since the goal of this contribution is to study the effect of EEFs on the reactivity of a solvated reaction complex, we started with an MD simulation devoid of any applied EEF, for the reactants (pyridine + methyl iodide) in a solvent box consisting of CH_3CN molecules. MD simulation leads to the expected conclusion that the solvent network within the box takes on a disordered geometry over the course of the simulation time and is in a constant flux: the overall dipole moment of the simulation box exhibits a great deal of random fluctuation (see Section S4 in the SI). In order to quantify the degree of disorder more directly, we determined the overall entropy of the system, which amounted to 10.0 kcal/mol·K.

Our QM/MM calculations reveal that even though the solvent molecules are not spatially ordered at any point throughout the MD simulation, still their collective effect on the Menshutkin reaction is significant. Whereas in the gas-phase, the reaction mechanism associated to this reaction is essentially concerted and exhibits an excessively high barrier of 27.4 kcal/mol (cf. Section S5 in the SI), in the presence of a solvent environment, the reaction turns effectively into a stepwise process. Thus, as shown in Fig. 2, the initial step in the mechanism involves the formation of a reactive cluster IM1 along the N---C---I axis. The second step involves the actual displacement reaction step.

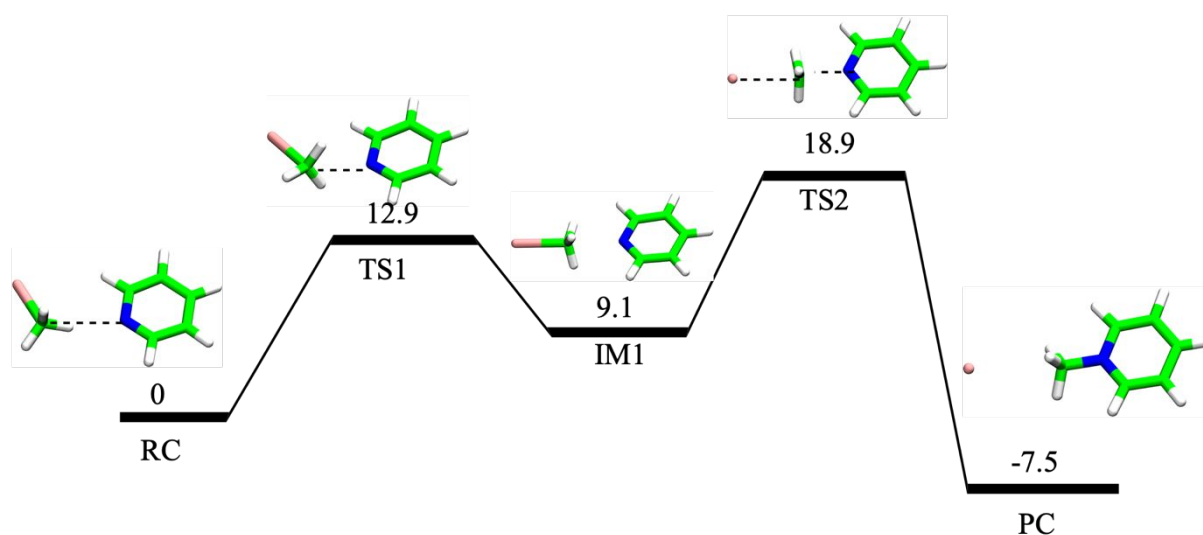


Figure 2. The potential energy surface (PES) associated to the Menshutkin reaction between pyridine and CH_3I in a field-free CH_3CN -solvent environment with its characteristic species indicated as follows: the reactant complex (RC), first transition state (TS1), intermediate (IM), second transition state (TS2) and product complex (PC). Energies (relative to RC) are shown in kcal/mol.

The effective kinetic barrier for the full process amounts to 18.9 kcal/mol (cf. Section S6 in the SI).³⁴ Hence, the field-free solvent environment unequivocally exerts a catalyzing effect.³⁵ For acetone and chloroform, a similar, though less pronounced, catalysis is observed (cf. Section S7 in the SI). To assess whether this catalysis imparted by the solvent environment is mainly electrostatic in nature, a QM/EF calculation was performed next to the QM/MM calculation (cf. Methodology Section). In a QM/EF calculation, a regular QM calculation is performed with the charge distribution associated to the environment taken into account by use of point charges of the solvent molecules. Performing such a calculation for the optimized QM/MM geometries for the critical species associated to the Menshutkin reaction leads to values that deviate 0.5-1.3 kcal/mol from the original QM/MM-barriers listed in Fig. 2 (cf. Table 1).

Table 1. Energies of the critical species, for the Menshutkin reaction in acetonitrile, calculated at QM/MM and QM/EF level of theory respectively (UB3LYP/B2+ZPE). The energy of the respective reactant complex was taken as the reference.

Species	QM/MM energy (kcal/mol)	QM/EF energy (kcal/mol)
RC	0.0	0.0
TS1	12.9	13.7
IM	9.1	10.2
TS2	18.9	19.4
PC	-7.5	-6.2

This finding confirms that the effect of the solvent on the main features of the reaction is indeed almost exclusively electrostatic in nature, or put differently, this indicates that *the solvent environment can be modeled quite accurately by use of a simple point-charge distribution*.

Using the TITAN code, we were able to quantify the net intrinsic electric field, exerted by the solvent molecules, inside the Menshutkin cavity, in the EEF-free reaction. We observe that this field changes dramatically throughout the cavity as the reaction proceeds, due to the dynamic reorganization of the surrounding disordered solvent network (Fig. 3).

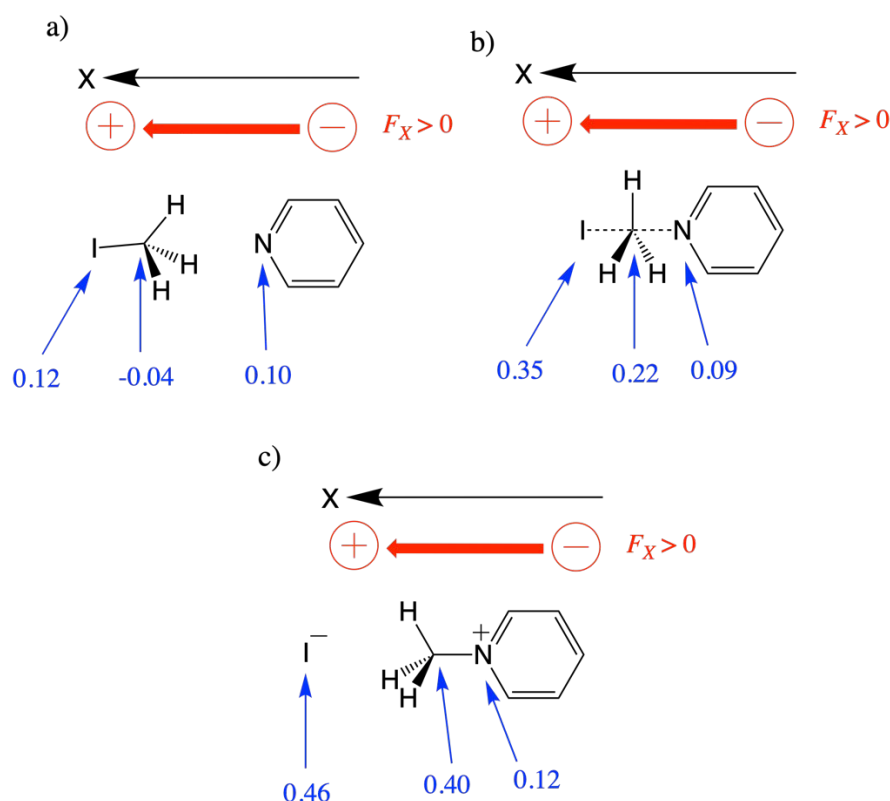


Figure 3. The component of the intrinsic electric field (in V/Å units) aligned with the reaction-axis (F_X) at a number of positions throughout the Menshutkin cavity in CH₃CN solvent for (a) the reactant geometry, (b) the TS2 geometry and (c) the product geometry.

In the reactant geometry, the magnitude of the EF component aligned with the reaction axis fluctuates significantly and takes on both positive and negative values, ranging from -0.04

up to +0.12 V/Å. In the TS2 geometry on the other hand, the electric field corresponds to a more or less uniform oriented field: F_X is consistently positive across the cavity and remains almost constant in magnitude. In the product geometry, the field strength of this emerging oriented intrinsic electric field increases even further. Note that the orientation of the observed oriented intrinsic-electric field matches exactly the field direction needed for catalysis of the Menshutkin reaction (cf. Fig. 1). As such, *the solvent itself effectively takes on the role of a catalyzing oriented EF throughout the reaction.*

Note that the appearance of a solvent-induced oriented electric field throughout the reaction is a logical consequence of the gradual polarization occurring within the Menshutkin complex, i.e. the emergence of a positive charge on the pyridine and a negative charge on the iodide moieties. In the proximity of the pyridine, the $\text{CN}^{\delta-}$ -moieties of the solvent molecules become increasingly directed toward the cavity; in the region around the iodine on the other hand, it is the $\text{CH}_3^{\delta+}$ -moieties of the molecules, which are increasingly directed toward the Menshutkin complex.

Note also that the differential electrostatic stabilization of the species along the PES by the electrostatic environment of the solvent described above is in fact the basis of the implicit solvent models ubiquitously used in quantum chemistry.³⁶

MD simulations of the solution under the influence of an EEF

To assess the effect of EEF application on the reagents and their solvent environment, MD simulations were subsequently performed in the presence of an electric field of incremental strength. For the acetonitrile solvent, the field strength was varied from 0.02 to 0.2 V/Å (0.02, 0.03, 0.035, 0.04, 0.05, 0.06, 0.07, 0.1, 0.15, and 0.2 V/Å). Fig. 4 shows the EEF effect on the solvent ordering, where Fig. 4a indicates the convention of the polarity of the applied EEF. The solvent distribution around the reactant complex (pyridine + methyl iodide) undergoes a steady

evolution from a disordered state, in the absence of an EEF, to an ordered/aligned one as the electric field strength increases. Fig. 4b shows the disordered solvent, whereas Fig. 4c displays the ordered solvent.

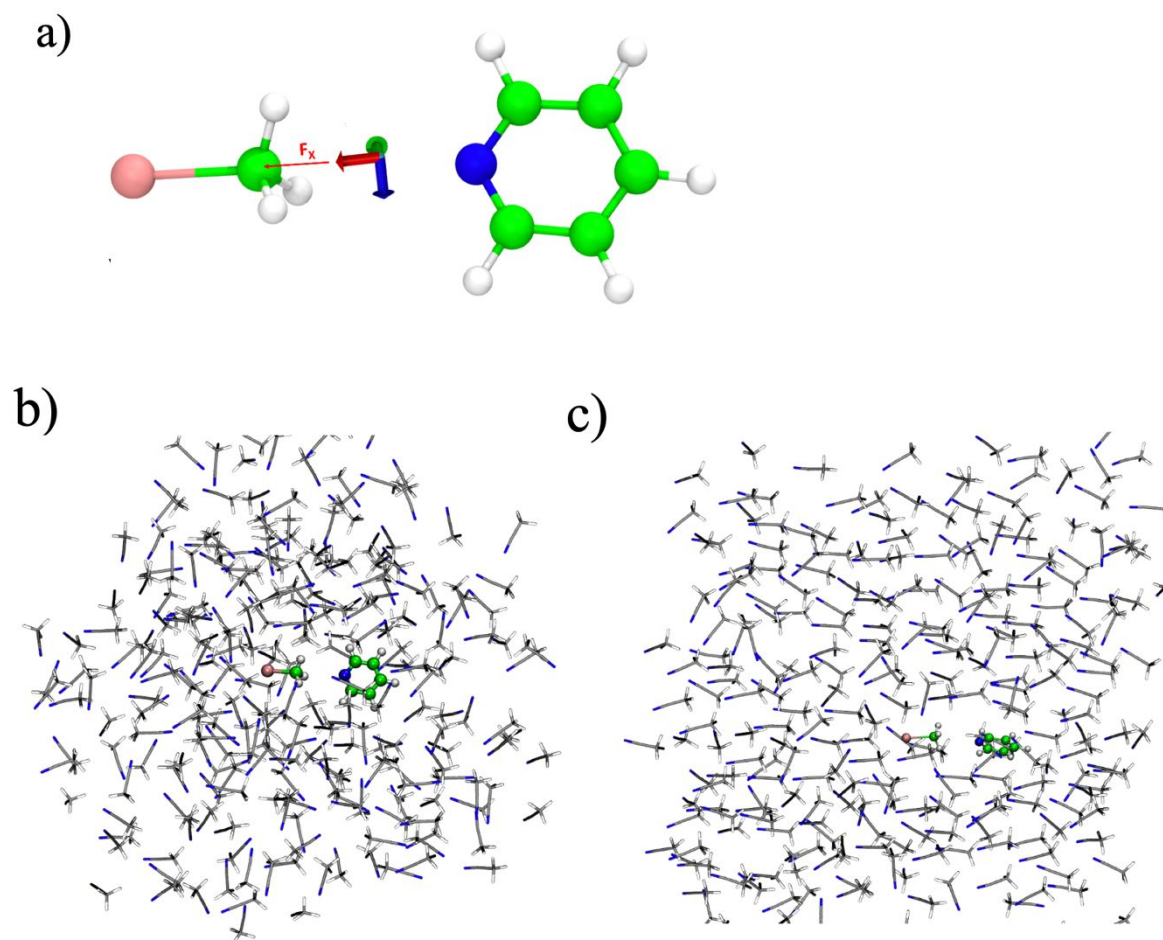


Figure 4. (a) The reaction system and the convention for a positive EEF applied in the study. (b) Disordered polar solvent (CH₃CN), which is obtained when the electric field strength is either zero or very small. (c) Significantly more ordered CH₃CN, which is obtained for a moderately strong electric field, i.e. $F_x > 0.1$ V/Å. In both parts (b) and (c), the reactants (pyridine---methyl iodide) are present in the middle of the simulation box.

Fig. 5 allows a visual inspection of the representative snapshots in the MD trajectories for different F_x values. It is seen that the solvent alignment increases with an increasing strength of F_x , and the global dipole moments of the solvent molecules (red arrows in Fig. 5) gradually

aligns with the field along the X axis. At $F_X \geq 0.15$ V/Å, almost all CH₃CN molecules are aligned, and so is the global dipole moment. Note that this finding, i.e. that external electric fields induce reorientation and ordering in a liquid phase, is in-agreement with previous experimental results,³⁷ and computational results by Evans,³⁸ Cassone *et al.*^{18,39} as well as by others.⁴⁰

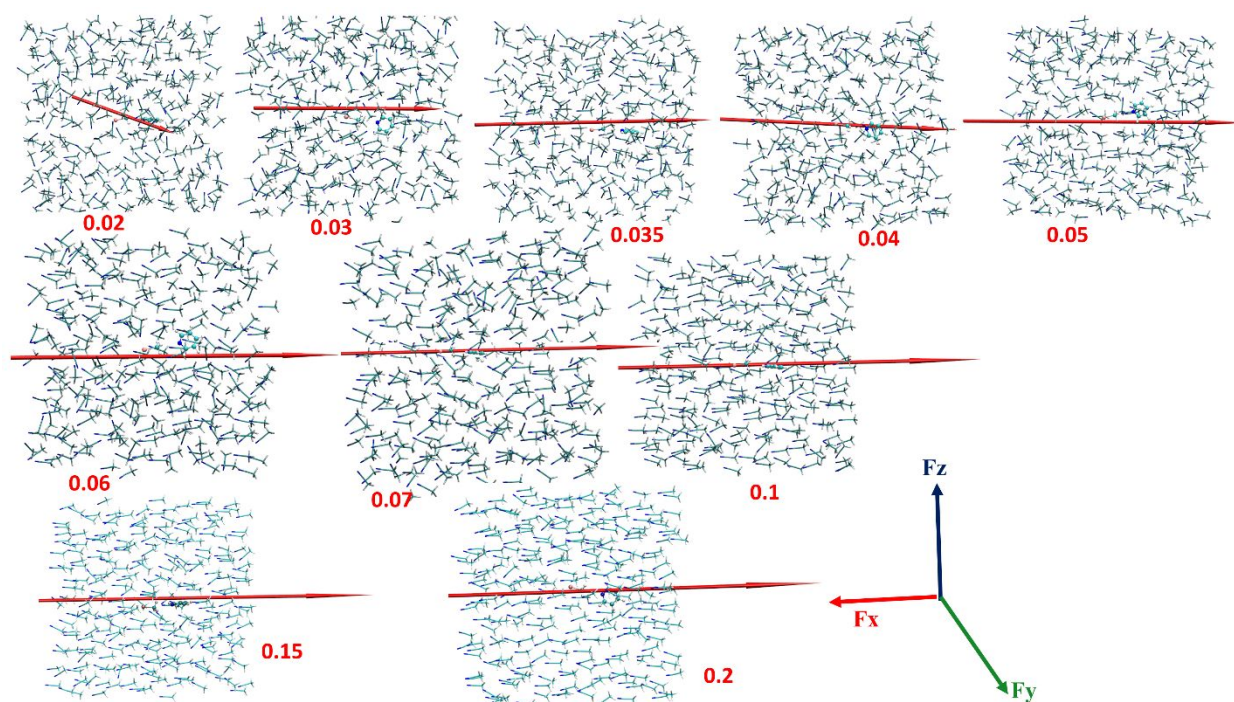


Figure 5. Representative snapshots from MD simulations of a CH₃CN solvent box at different positive F_X values (in V/Å), which are noted in red, by each individual snapshot. The convention for a positive field is the same as shown in Fig. 1a. The red arrows drawn in each frame show the direction of the global dipole moment of the solvent ensemble in the box (in Debye units); the convention for the dipole moment vector assigns the head of the arrow as the positive pole.

To further elaborate on the effect of EEFs on the solvent ensemble, we also calculated the evolution of the entropy of the solvent as well as the net dipole moment of the system, as

the EEF strength increases (Fig. 6a and 6b respectively). It is quite clear that the order, as well as the global dipole-moment of the ensemble increases gradually with increasing strength of the applied EEF. It is however important to note that this initial rise is followed by a levelling off, for both the solvent's entropy and dipole moment. As such, there seems to be a critical field strength above which the solution is no longer able to change its structure in response to further increase of the EEF. This observation can be straightforwardly rationalized by realizing that the mechanism through which the solvent responds to the application of an EEF is a collective orientation of the solvent molecules: *at some point the solution will have become perfectly aligned/organized so that the dipole moment can no longer be increased this way, i.e. dielectric saturation of the solution has been reached.*⁴¹

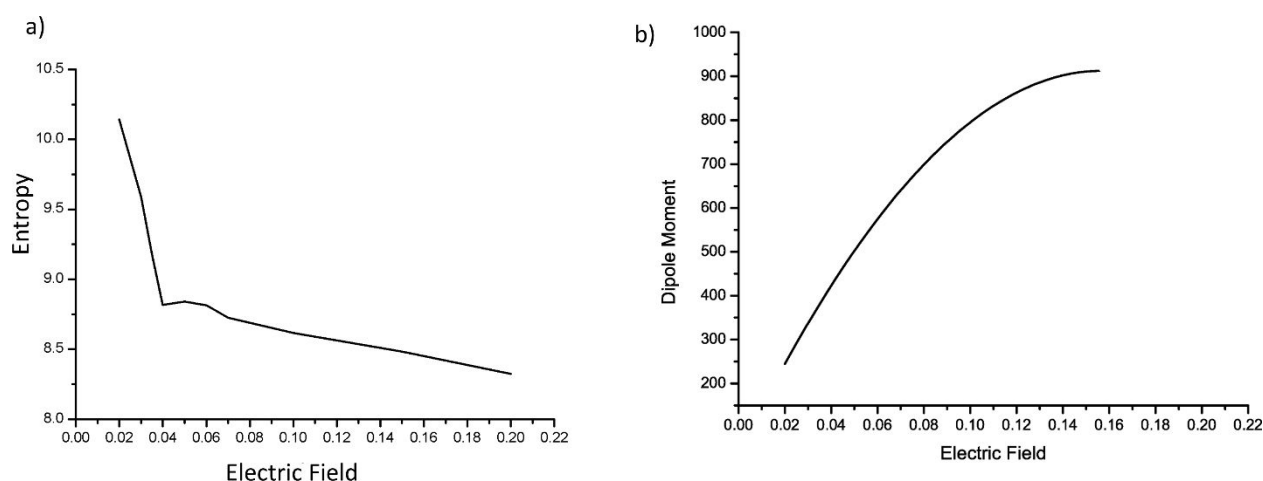


Figure 6. Smoothed plots of (a) the solvent entropy (in kcal/mol K), vs. F_X (in V/Å), and (b) the dipole moment (in Debye units) of the CH₃CN solvent system vs. F_X (in V/Å).

Further evidence for this reasoning can be found in the observation that for smaller field strengths, e.g. 0.02 V/Å, the standard deviation in dipole moment is ± 50 D (Debye) while for stronger EEFs the standard deviation drops to ± 10 D (see the root mean square atomic fluctuations in Section S8 and the evolution of the overall dipole moment of the simulation box in Section S9 in the Supporting Information), i.e. at some point a maximal alignment of the individual dipoles is reached. In other words, *one can say that the EEF acts as a tweezer, which*

both aligns the solvent molecules (see also Fig. 2c and 6) and keeps them from fluctuating due to thermal motion.⁴²

For the other solvents tested, i.e. acetone and chloroform, similar qualitative trends are observed, but the calculated critical electric field strength at which perfect solvent alignment, i.e. dielectric saturation, is reached, is shifted upwards as the polarity of the solvent decreases (already for acetone, $F_{X,critical} > 1 \text{ V/\AA}$). This behavior can be connected to the lower individual dipole moments of the CHCl_3 and CH_3COCH_3 molecules: *a higher external field strength is needed to overcome thermal fluctuations and produce sufficient stabilization of the perfectly aligned orientation*. This is also reflected in the observation that the growth of the global dipole moment as a function of increasing field strength is reduced for these solvents (cf. Section S9 of the SI).

Note that the use of a non-polarizable force-field in our MD overestimates the calculated critical field strength value; in reality, the dipole moment of the individual solvent molecules will increase as the EEf-strength grows,^{1b} which will facilitate the orientation of the molecules.

How does the solvent ordering affect the observed catalysis?

One can logically expect that the emergence of solvent ordering under the influence of an external electric field will significantly impact the solvent-induced catalysis. Recall that the solvent-induced catalysis is essentially a local and dynamic process: the solvent molecules adjacent to the cavity reorganize as the reaction proceeds so that the reacting complex is optimally stabilized at each step throughout the transformation from reactant to product. For the Menshutkin reaction, this essentially means that the solvent environment effectively gives rise to a catalyzing oriented EF of gradually increasing field strength (Fig. 3). *Fixing the orientation of the individual molecules by the EEf will obviously hamper the ability of the solvent to produce such a catalyzing oriented field.*

Additionally, the collective ordering of the many layers of dipoles, making up the liquid phase, induces a global static electric field in its own right (Fig. 6). As can be seen from the sketch in Fig. 7, the direction of the resulting solvent-induced electric field points in the opposite direction of the applied EEF.

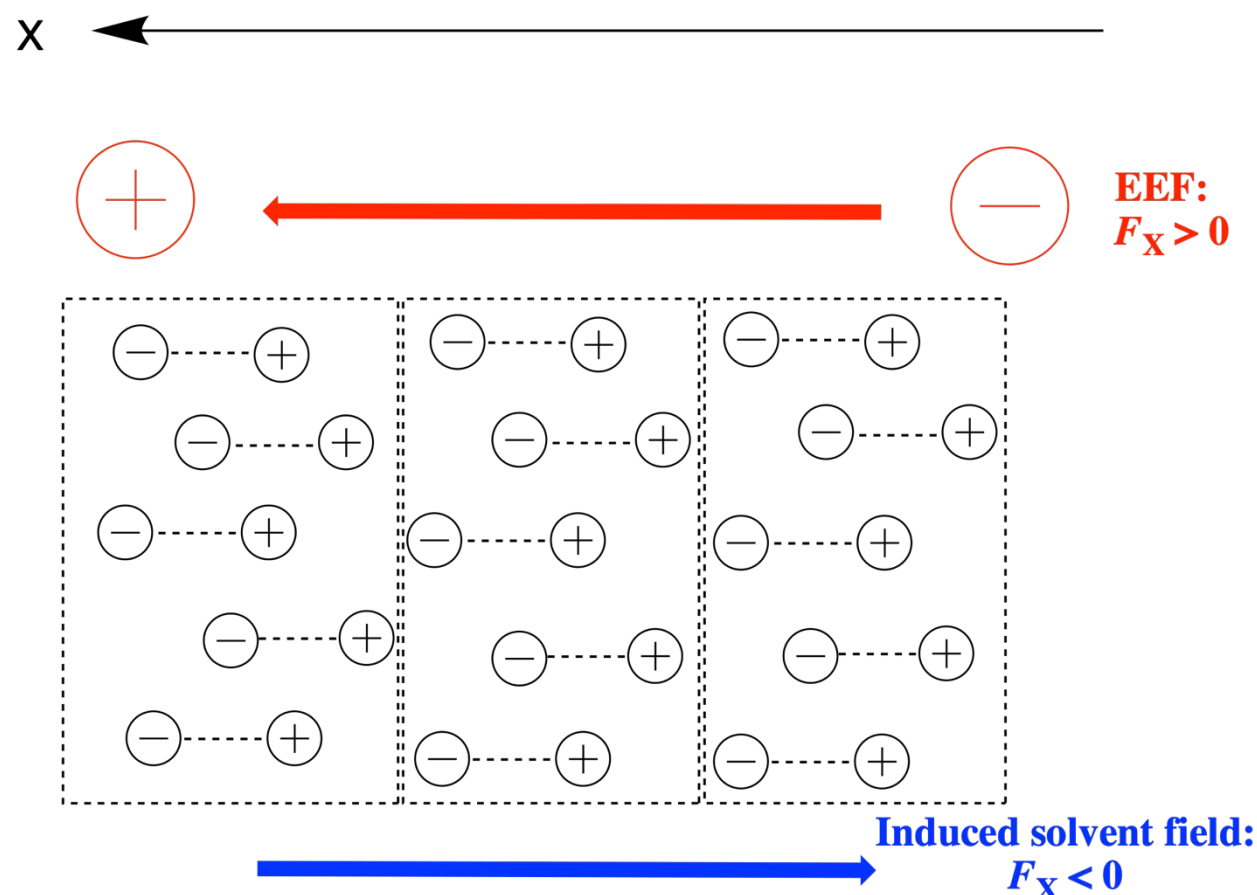


Figure 7. An applied EEF ($F_X > 0$) induces the collective orientation of the dipoles (corresponding to the individual solvent molecules) depicted. The point charge distribution associated to these dipoles in their turn lead to the emergence of an ‘induced solvent field’ ($F_X < 0$), which counteracts the originally applied EEF. Idealized solvent layers are denoted by the dashed boxes. Note the relative displacement of the dipoles within each individual solvent layer, which minimizes the intralayer dipole-dipole repulsion.

The collective orientation of the solvent molecules will thus result in a partial – or complete – cancellation of the effect of the external field, i.e. the field is “screened”. The extent

of this screening can be expected to depend mainly on the polarity of the solvent i.e. the magnitude of the “point charges” on opposite sides of the molecular axis: for strongly polar solvents like acetonitrile, the screening will be almost complete (up to the critical field strength at which perfect orientation is reached, *vide supra*), for less polar solvents, the organization and screening will be much less pronounced. Another factor which will contribute to the extent of screening is the dimensions of the reaction medium, i.e. the number of layers of solvent molecules which can align collectively.⁴³

Our computational results confirm this point-of-view. We performed QM/MM calculations under the influence of a strong oriented external electric field (OEEF) of 0.5 V/Å, for the different solvents (cf. Fig. 8 for the geometry of the simulated system in the case of CH₃CN solvent). The EEF was incorporated into the calculation through circular charged capacitor plates, generated with TITAN (cf. Methodology Section; the plates were included as “inactive atoms” in the MM region). Furthermore, we kept the outer layers of the solvent box frozen in the EEF-aligned orientation obtained from the MD simulation (by making them inactive as well), so as to enforce the shape of the simulation box and thus preventing unphysical separation of the solvent molecules during the simulation (cf. Methodology Section). The charged plates were placed on opposite sides of the solvent box along the X-direction so that the OEEF aligns with the reaction axis of the Menshutkin complex. Note however that this choice is in fact trivial since the reaction axis itself (together with all the solvent molecules) will realign themselves in case the direction of the OEEF is changed, due to the tweezing behavior of the field (cf. Section S10 in the Supporting Information).

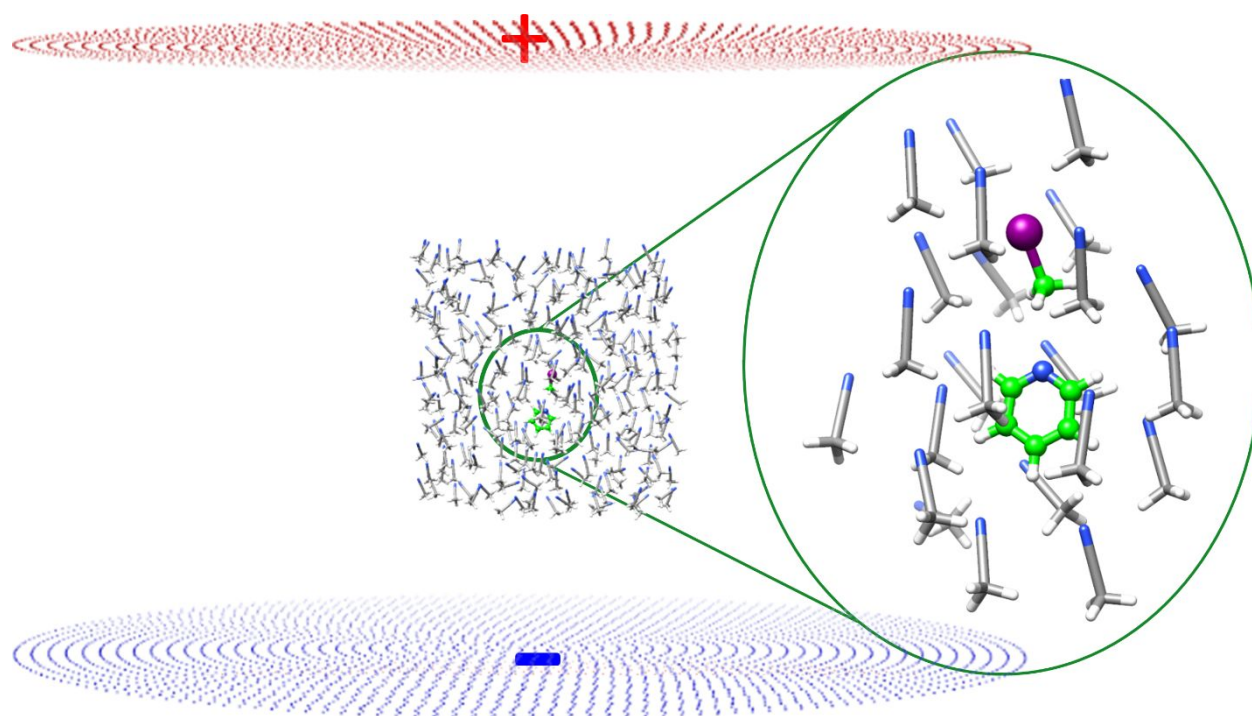


Figure 8. The geometry of the solvated system (in the case of CH_3CN) for the QM/MM calculation under the influence of the EEF. Only the inner MM atoms within $6\text{\AA}$ of reactant were relaxed (i.e. allowed to respond to the EEF), while the rest was kept frozen in the aligned geometry coming out of the MD simulation in the presence of an EEF. Note the solvent orientation around the reaction system in the enlarged view on the right. The red and blue objects are respectively the positively and negatively charged circular plates generated with the help of the TITAN code.

The calculated PES for the Menshutkin reaction in acetonitrile, in the presence of the OEEF, is shown in Fig. 9. Two conclusions can be drawn from this Figure; (a) the barrier associated to the first reaction step has completely disappeared now, and (b) the overall barrier for the reaction has dropped significantly.

The disappearance of the barrier for reactive complex formation can be connected to the EEF-induced orientation of the reactants: *the EEF acts as a tweezer that enforces the correct alignment of the reactants, required for the displacement step*, by default. In addition to the OEEF-induced disappearance of the first barrier, one can also observe that the barrier for the overall process has been lowered by the OEEF to 8.3 kcal/mol . Furthermore, no product cluster is formed anymore: once the TS has been crossed, the energy continues to decrease as the charged products separate, i.e. one cannot observe a local minimum on the product side of the

PES. Overall, we can unequivocally conclude that the applied field imparts significant catalysis on the solvated system (relative to the barriers of 27.4 and 18.9 kcal/mol in the gas phase and EEF-free solvent, respectively).

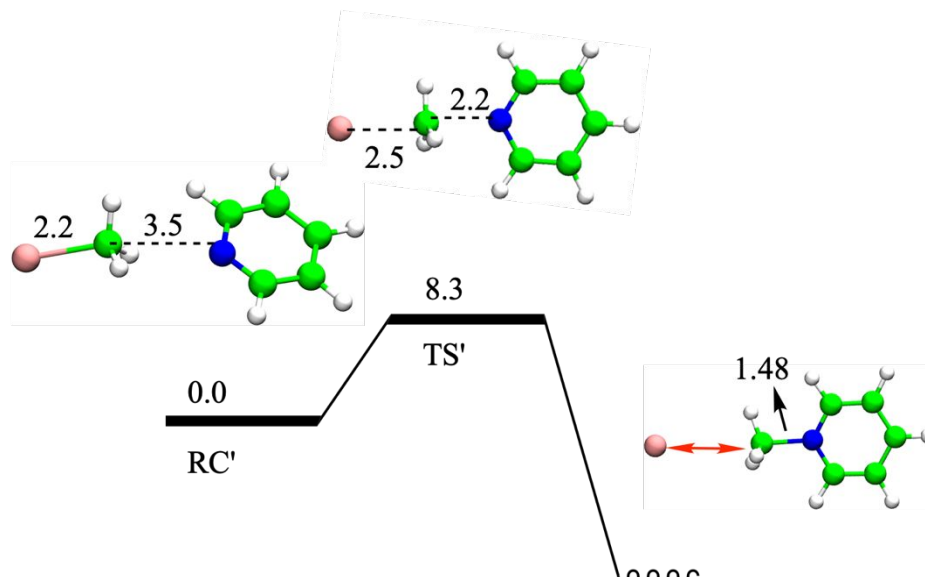


Figure 9. The potential energy surface (PES) associated to the Menshutkin reaction between pyridine and CH_3I in a CH_3CN solvent environment exposed to an EEF of $0.5 \text{ V/\AA}$ with its characteristic species indicated: the reactant complex (RC'), the transition state (TS'). Since no local minimum is observed on the product-side of the PES at the considered field strength, the product is represented by a wavy line. Energies are shown in kcal/mol.

For acetone and chloroform, the barriers obtained in the presence of the OEEF are – as expected – lower, they amount to 8.1 kcal/mol and 6.9 kcal/mol respectively, indicating a somewhat lower extent of screening (cf. Section S11 in the Supporting Information).

Analyzing the in-cavity field strength, induced by the organized solvent molecules alone, i.e. without taking the charged plates into account, reveals that the enhanced catalysis for chloroform compared to acetonitrile and acetone can indeed be connected to a reduced screening (Fig. 10). For all solvents, the oriented electric field which was present in the field-free solvent situation (cf. Fig. 2) has been upended. For chloroform however, only a very mild

(global) counteracting field has emerged, whereas for acetonitrile, this counteracting field is significant, and the direction of the original solvent-induced oriented field has essentially been flipped.

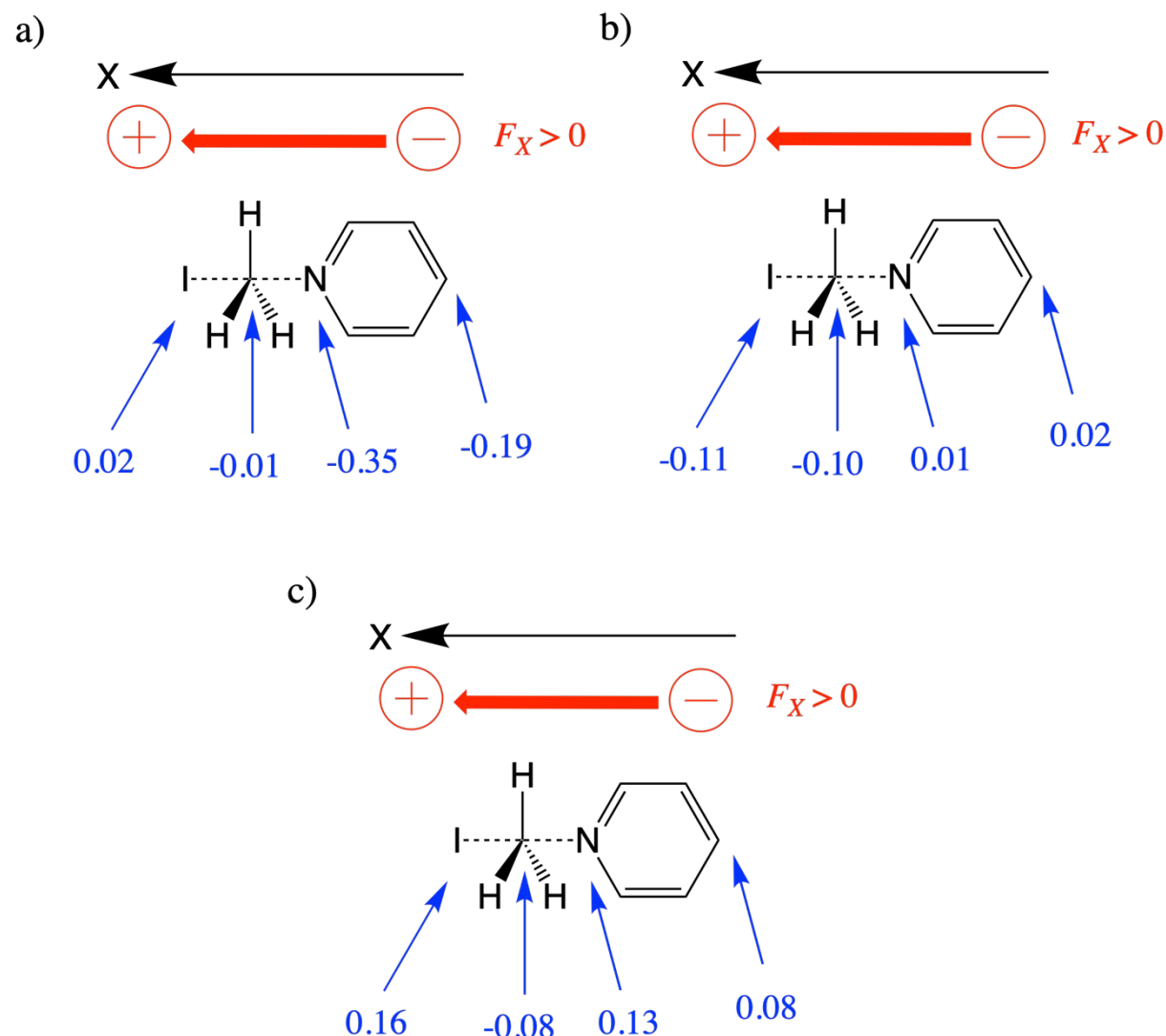


Figure 10. The component of the electric field ($0.5 \text{ V/\AA}$) aligned with the reaction-axis (F_X) exerted by the organized solvent at a number of positions throughout the Menshutkin cavity in the TS geometry for (a) acetonitrile, (b) acetone and (c) chloroform.

Note that solvent polarization in response to the exerted EEF – which was not considered in our QM/MM calculations, cf. Methodology Section – can generally be expected to increase the strength of the solvent-induced electric field, i.e. enhance the screening.

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3 However, once the solvent molecules start to polarize, so will the Menshutkin complex. Since
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5 polarization of the reaction complex was exactly what caused EEFs to induce catalysis in the
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7 gas-phase situation (cf. Fig. 1), catalysis will inevitably emerge as soon as charge separation
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9 sets in within the liquid phase, i.e. as soon as the OEEF exceeds the counteracting field
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11 emerging from the collective reorientation of the (unpolarized) solvent molecules.
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15 Finally, we also verified the counteracting nature of the aligned solvent, in its own right,
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17 on the Menshutkin reaction. When we extract the geometries of the different critical species
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19 coming out of the QM/MM calculation described above, and perform a QM/EF calculation on
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21 the system, without the OEEF delivered by the charged plates, and with only the point charges
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23 associated to the organized solvent taken into account, we find that the energy of the system
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25 increases monotonically throughout the entire reaction, i.e. *the solvent-induced electric field, in*
26
27 *its own right, indeed inhibits the reaction (and product formation) altogether.*
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31 The slopes of the resulting monotonically increasing energy curves are solvent-polarity
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33 dependent and thus reflective of the magnitude of the generated solvent-induced electric field,
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35 in line with what one would expect from the discussion above. Thus, for acetonitrile, we find
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37 that the energy of the original TS geometry in the presence of the solvent-induced field amounts
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39 to 21.6 kcal/mol. For acetone, this value decreases to 19.5 kcal/mol, and for chloroform, the
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41 corresponding energy amounts to barely 13.0 kcal/mol.
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45 Finally, let us re-consider whether the experimental data available in the literature also
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47 support the expectations outlined above. As indicated already in the Introduction, STM
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49 experiments in the presence of nonpolar solvents indeed lead to catalysis in a similar way as
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51 one would expect in the gas-phase.^{3,4,5} Furthermore, as tested by Venkataraman *et al.*, upon
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53 replacing the nonpolar solvents by polar ones, the catalytic activity of the EEF decreases
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55 steadily when the field strength is kept constant.⁶ Both of these observations are in agreement
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57 with what one would expect based on our pictorial analysis.
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Generality of the Model of OEEF Effects in a Solvent

Despite the fact that the preceding discussion focused on a single reaction, our model and conclusions are generalizable, since the catalysis exerted by electric fields follows a universal paradigm. As discussed in our previous studies,^{1a-b,2a} and briefly mentioned in the introduction of this paper, electric fields impart catalysis by disproportionally stabilizing the so-called “VB charge transfer states (CTS)”, which are responsible for the charge flow from one reactant to the other and thus for the emergence of polarization within the reaction complex.

Generally, these CTS states do not contribute in an equal way to all the species along the PES, some stationary points are more affected by them than others, and it is this differential susceptibility along the PES which causes the catalysis, i.e. the reduction of the global barrier associated to the process.

For the Menshutkin reaction, the CTS contribution (which – as indicated above – is related to the extent of polarization, i.e. the rise in the magnitude of the dipole moment of the species) increases throughout the entire reaction course, so that both the TS and PC are stabilized by the field relative to the RC.

Other reactions can be discussed in a similar manner, since all reactions possess CTSs. For example, in the Diels-Alder reaction of cyclopentadiene and maleic anhydride,^{1,2a} and in oxidative addition reactions,^{2b} e.g. the reaction between $\text{P}(\text{PH}_3)_2$ and CH_3X , it is mainly the TS which is disproportionately stabilized by the field. In any case, since the energy difference between RC and TS decreases for all these reactions, the kinetics will improve under the influence of an electric field, i.e. catalysis sets in.

From our current analysis, one can conclude that the OEEF enhancement of the CTS effect transpires easily in the presence of apolar or mildly polar solvents, where solvent organization by the field will not be so significant (e.g., see CHCl_3 and to a lesser extent

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3 acetone), but may also take place in strongly polar solvents such as CH_3CN , as soon as the
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5 OEEF overcomes the screening.
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8 Whether or not the reactants are pre-organized (or not oriented in the right way for
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10 reaction to occur) will only play a minor role in the catalysis: polar reactants will generally
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12 become oriented in the correct direction for reaction to occur, i.e. the negative pole of one
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14 reagent will be directed towards the positive pole of the other in the field direction (cf. the
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16 Menshutkin complex in Fig. 1). For apolar reactants, the EEF-mediated pre-organization may
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18 not always be favorable, but due to the small dipole moment of these compounds at the onset
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20 of the reaction, reorientation will be facile. As such, the energy lowering of the TS, caused by
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22 the improved mixing of the CTSs as the reaction proceeds, will still dominate the shape of the
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24 PES and catalysis will emerge.
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30 31 **Conclusions**

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33 To summarize, let us consider the overall picture that emerges from our analysis. Our
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35 calculations indicate that in the absence of an EEF, solvents take on a catalyzing function by
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37 gradually developing an oriented intrinsic-electric field, facilitating the “electron flow”
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39 associated to the transformation from reactants to products. Applying an OEEF causes the
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41 solvent to undergo organization, i.e. the solvent molecules gradually align with the applied field
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43 as the field strength increases. This collective organization has a major impact on the catalysis
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45 provided by the solvent. On one hand, it restricts the positioning of the solvent molecules
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47 around the reagent-cavity, thus subduing the emergence of the (local) solvent-induced uniform
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49 oriented field. On the other hand, it induces a global electric field pointing in the opposite
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51 direction of the applied EEF. The combination of these two intertwined effects leads to (partial
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53 or complete) screening of the OEEF, with the extent of screening being proportional to the
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55 polarity/polarizability of the solvent.
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Nevertheless, even though the applied OEEF is clearly attenuated by the solvent environment, *we observe for each of the solvents tested that catalysis inevitably emerges once the OEEF exceeds the opposing field of the organizing solvent and polarization of the Menshutkin complex sets in.* Overall, our analysis provides a lucid and pictorial interpretation of the behavior of solutions in the presence of OEEFs and indicates that OEEF-mediated catalysis should in principle be feasible in bulk setups, especially for nonpolar and mildly polar solvents. As shown above, these conclusions and the eventual emergence of catalysis are not particular to the Menshutkin reaction, but can be generalized to other reactions as well.

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at <http://pubs.acs.org>

Detailed information concerning the molecular dynamics simulations, technical motivation for the subdivision of the solvent box, estimation of the entropic contribution to the observed catalysis, evolution of the overall dipole moment of the simulation box over time ($F_X = 0$), gas-phase reaction profile of the Menshutkin reaction, effective kinetic barrier for the full Menshutkin reaction in acetonitrile depicted in Fig. 2 of the main text, potential energy surface associated to the Menshutkin reaction between pyridine and CH_3I in a field-free acetone and chloroform solvent, root mean square deviation of the solvent at three different EEF field strengths, evolution of the overall dipole moment of the simulation box over time ($F_X \neq 0$), realignment of the solvent and reaction axis upon application of an OEEF aligned with the y-axis, simulation results for acetone and chloroform ($F_X \neq 0$), geometries for all the QM systems.

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Notes

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TOC graphic

